TANAY AGRAWAL

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EDUCATION

The Harker School Senior, Class of 2027
GPA: 4.0/4.0 (UW), 4.53 (W) *SAT: 1570/1600*

- Relevant Coursework: AP Computer Science A, Neural Nets, Compilers, AP Calculus BC, AP Spanish Language, AP United States History, AP World History, AP Microeconomics, AP Macroeconomics, AP Statistics, AP Physics C, AP Psychology.

RESEARCH

Independent Researcher | **arxiv.org/abs/2609.25012** October 2025 - Present
A Computational Approach to Measuring Semantic Change in Sanskrit Literature

- Sole author of a peer-reviewed paper accepted at an international conference (AACL-IJCNLP SRW 2026). Published in-depth blog series on website (tanayagrawal.xyz/blog).
- Constructed linguistically informed pipeline to measure semantic drift using word embeddings on Sanskrit corpora. Performed sandhi splitting and lemmatization to train novel models. Achieved 90.4% accuracy.

Host & Founder, Language of Resistance | **languageofresistance.org** May 2025 - Present

- Podcast series with leading scholars, including a Pulitzer Prize-winning poet, investigating protest language across different movements and cultures.
- Analyzed 1,886 translations of effective slogans across 54 languages using Wikidata to explore shared semantic and linguistic patterns.

Speech Processing Scholar, COSMOS UCLA Summer 2026

- Studied college-level signal processing and auditory phonetics under UCLA faculty.
- Built physics-based deepfake speech detector using glottal, vocal tract, and room features; presented a research poster to peers, professors, and other visitors.

Linguistics Intern, Science Internship Program (SIP), UC Santa Cruz Summer 2025

- Conducted novel linguistic research under a PhD student on verb-phrase ellipsis in Turkish, Korean, and Mongolian.
- Performed daily elicitations and fieldwork with native speakers to build a typology across the Altaic language family. Presented to UCSC Linguistics Department.

EXPERIENCE

Co-Founder, Bay2BayHacks | **bay2bayhacks.xyz** May 2025 - Present

- Organized global AI hackathon (540+ participants); secured prizes worth $57K+ in sponsorships from leading companies including Jane Street and recruited expert judges from Tesla, Uber, and Microsoft.

Vice President, HarkerDev | **dev.harker.org** December 2024 - Present

- One of thirteen students selected for a student-led development club for the betterment of The Harker School community. Lead all operations and technical decisions for active projects.
- Built and deployed school parking management app used daily by 400+ students and faculty members; coordinated with administration to ship a production system.
- Other projects: built the Associated Student Body (ASB) website for the 2026-2027 school year and reworked the GPA Calculator tool to better serve the school population.

Language Tech Intern, Chauwk | **tiny.cc/github-chauwk** Summer 2024

- Built a RAG-based platform for 23 low-resource South Asian languages to make government schemes and employment leads accessible to blue-collar workers.
- Enabled native-language (audio and text) support via machine translation and automatic speech recognition.

Founder, SportsAI | **sportsai.us** July 2023 - June 2025

- Conceived AI-powered sports coaching system using computer vision to analyze player performance and provide real-time feedback from video.
- Developed product, vision models, and business strategy to pitch to potential investors.

HONORS

National Merit Scholarship Semifinalist	*September 2026*
International Linguistics Challenge (ILC) Finalist, Bronze Honorable Mention	*March 2026*
ILC Exceptional Distinction Award (one of three in United States)	*January 2026*
The President's Volunteer Service Gold Award (2x)	*2024, 2025*
PRAGNYA 2024 NeuroAlly Leader of the Year	*November 2024*
DECA Regional 3rd Place Winner, Top 10 State Finalist (International Business Plan)	*April 2024*
National Youth Advocacy Corps Fellow (Graduated with Distinction)	*2024-2026*